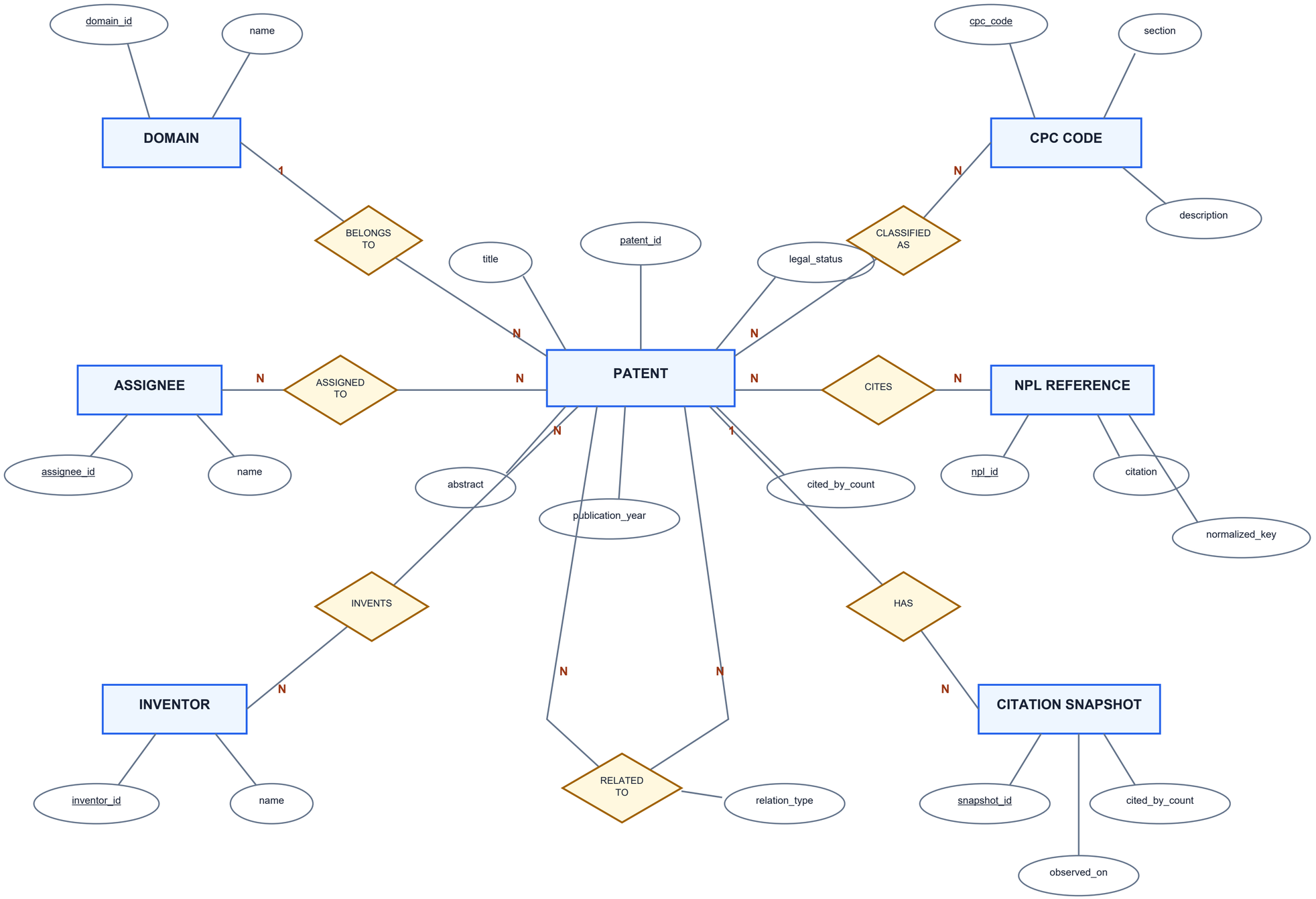
Patent Intelligence Platform — ER Diagram, Sheet 1 of 2

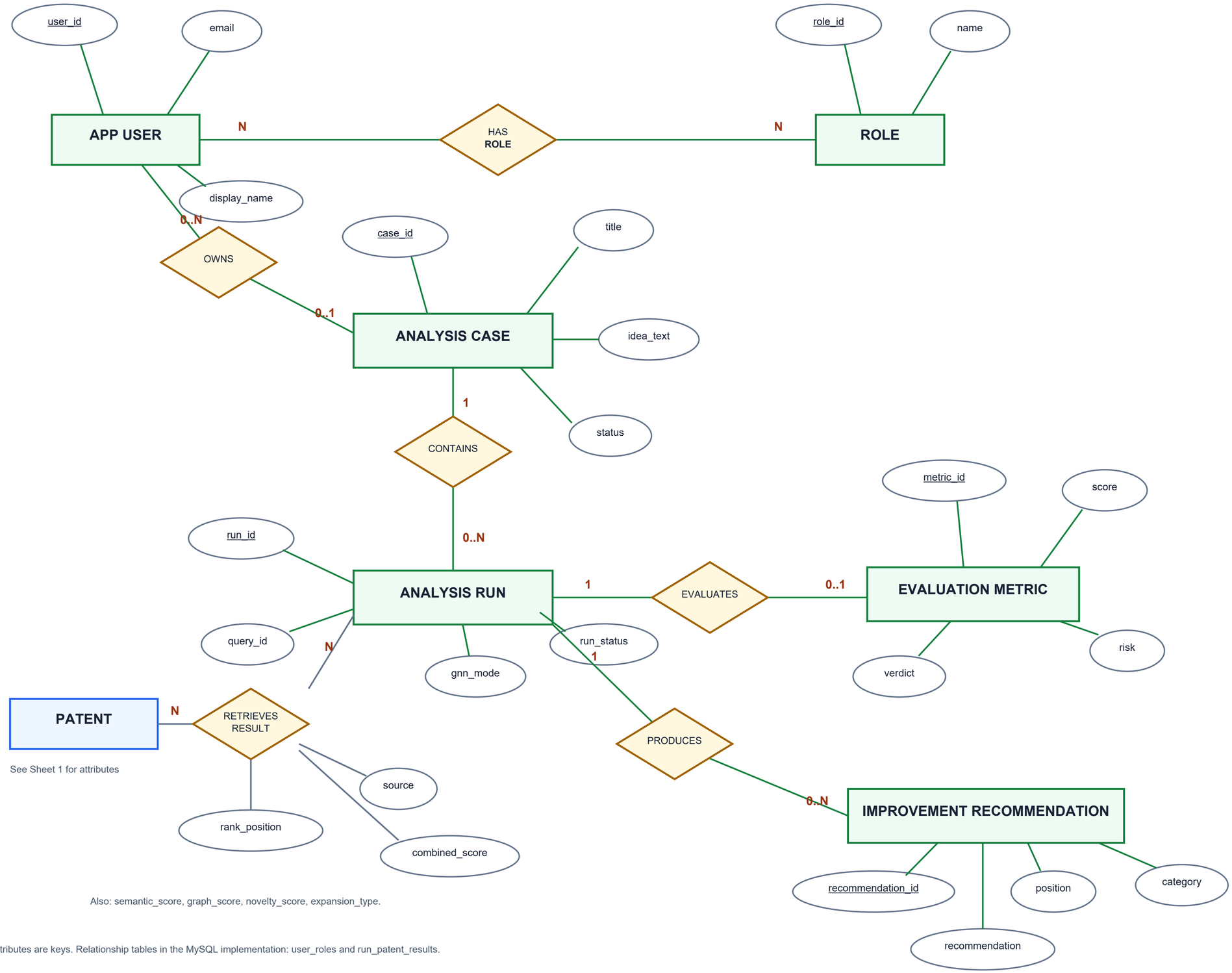
Patent knowledge, people, classifications, citations, and family relationships · Chen notation



Underlined attributes are keys. N:N relationships are implemented by junction tables: patent_assignees, patent_inventors, patent_cpc_codes, patent_npl_references, and patent_families.

Patent Intelligence Platform — ER Diagram, Sheet 2 of 2

User access, saved analysis cases, pipeline runs, evaluation metrics, and recommendations · Chen notation



See Sheet 1 for attributes

Also: semantic_score, graph_score, novelty_score, expansion_type.

Underlined attributes are keys. Relationship tables in the MySQL implementation: user_roles and run_patent_results.

PATENT INTELLIGENCE PLATFORM RELATIONAL SCHEMA

Each table appears once. Underlined attributes form the primary key. Arrows point from FK columns to their referenced PK columns.

Matching small numbers identify the two ends of each continuous arrow. Small jumps indicate crossings, not connections.

DOMAINS

domain_id	name

PATENTS

patent_id	title	abstract	publication_year	legal_status	jurisdiction	cites_patent_count	family_size	cited_by_patent_count	url	domain_id	created_at	updated_at

PATENT_DOMAINS

patent_id	domain_id

ASSIGNEES

assignee_id	name

INVENTORS

inventor_id	name

CPC_CODES

cpc_code	section	description

NPL_REFERENCES

npl_id	citation	normalized_key

PATENT_ASSIGNEES

patent_id	assignee_id

PATENT_INVENTORS

patent_id	inventor_id

PATENT_CPC_CODES

patent_id	cpc_code

PATENT_NPL_REFERENCES

patent_id	npl_id

PATENT_FAMILIES

patent_id	related_patent_id	relation_type

CITATION_SNAPSHOTS

snapshot_id	patent_id	cited_by_count	observed_on	created_at

APP_USERS

user_id	email	display_name	password_hash	is_active	created_at	updated_at

ROLES

role_id	name

USER_ROLES

user_id	role_id

ANALYSIS_CASES

case_id	owner_user_id	title	idea_text	status	created_at	updated_at

ANALYSIS_RUNS

run_id	case_id	query_id	idea_text	gnn_mode	top_k	run_status	started_at	completed_at	pipeline_payload	improvement_payload

RUN_PATENT_RESULTS

run_id	patent_id	rank_position	source	expansion_type	semantic_score	graph_score	combined_score	novelty_score

EVALUATION_METRICS

metric_id	run_id	patentability_score	risk	verdict	payload	created_at

IMPROVEMENT_RECOMMENDATIONS

recommendation_id	run_id	position	category	recommendation	rationale

AUTH_SESSIONS

token_hash	user_id	expires_at	created_at

AUTH_RATE_LIMITS

bucket_key	attempts	window_started

Independent throttle counters; no foreign keys.

Blue: patent references | Purple: catalog references | Green: account/role references | Ochre: analysis references

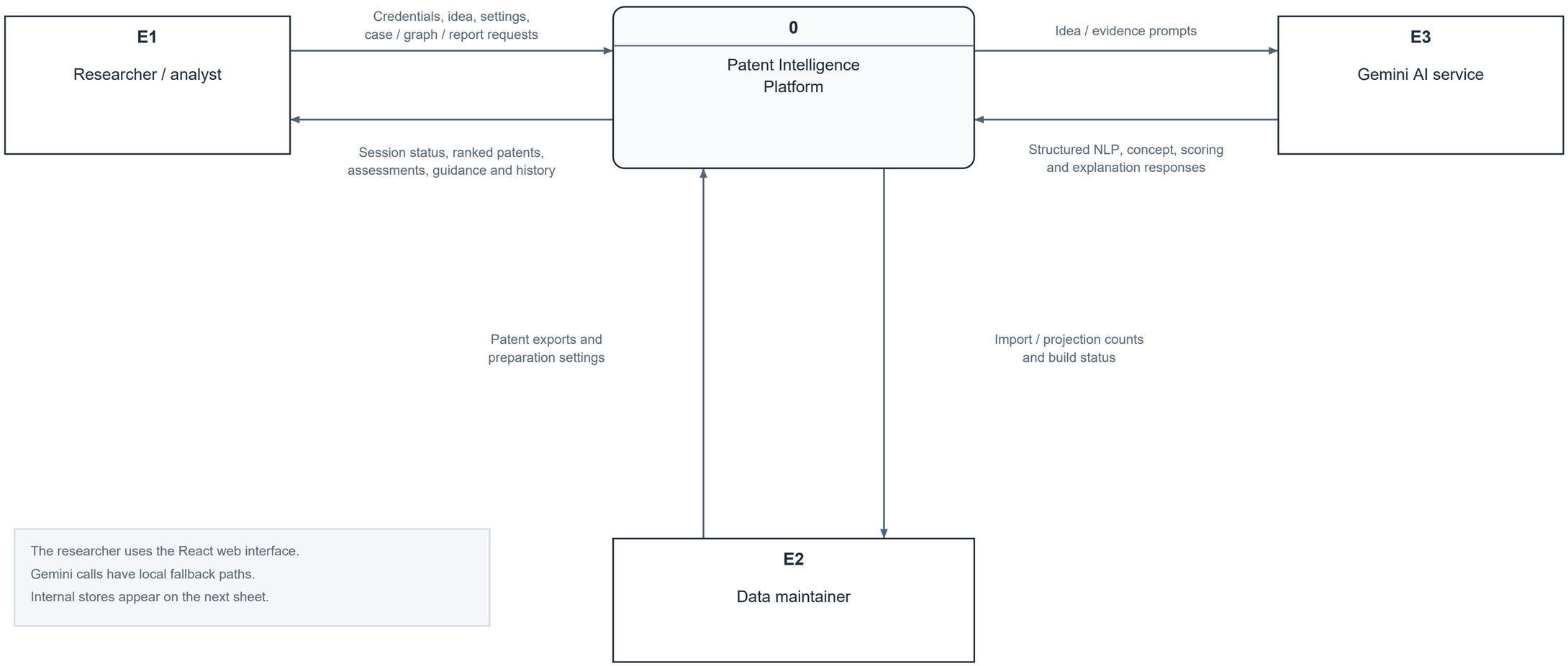
23 tables, 101 attributes, 23 foreign keys. Every arrow is continuous between actual columns; number tags are not attributes.

All underlined attributes in a relation together form its primary key. Source: database/sql/01_schema.sql and 05_auth.sql.

PATENT INTELLIGENCE PLATFORM

Level 0 - System context

System boundary includes the web application, analysis services and offline data preparation.



The researcher uses the React web interface.
Gemini calls have local fallback paths.
Internal stores appear on the next sheet.

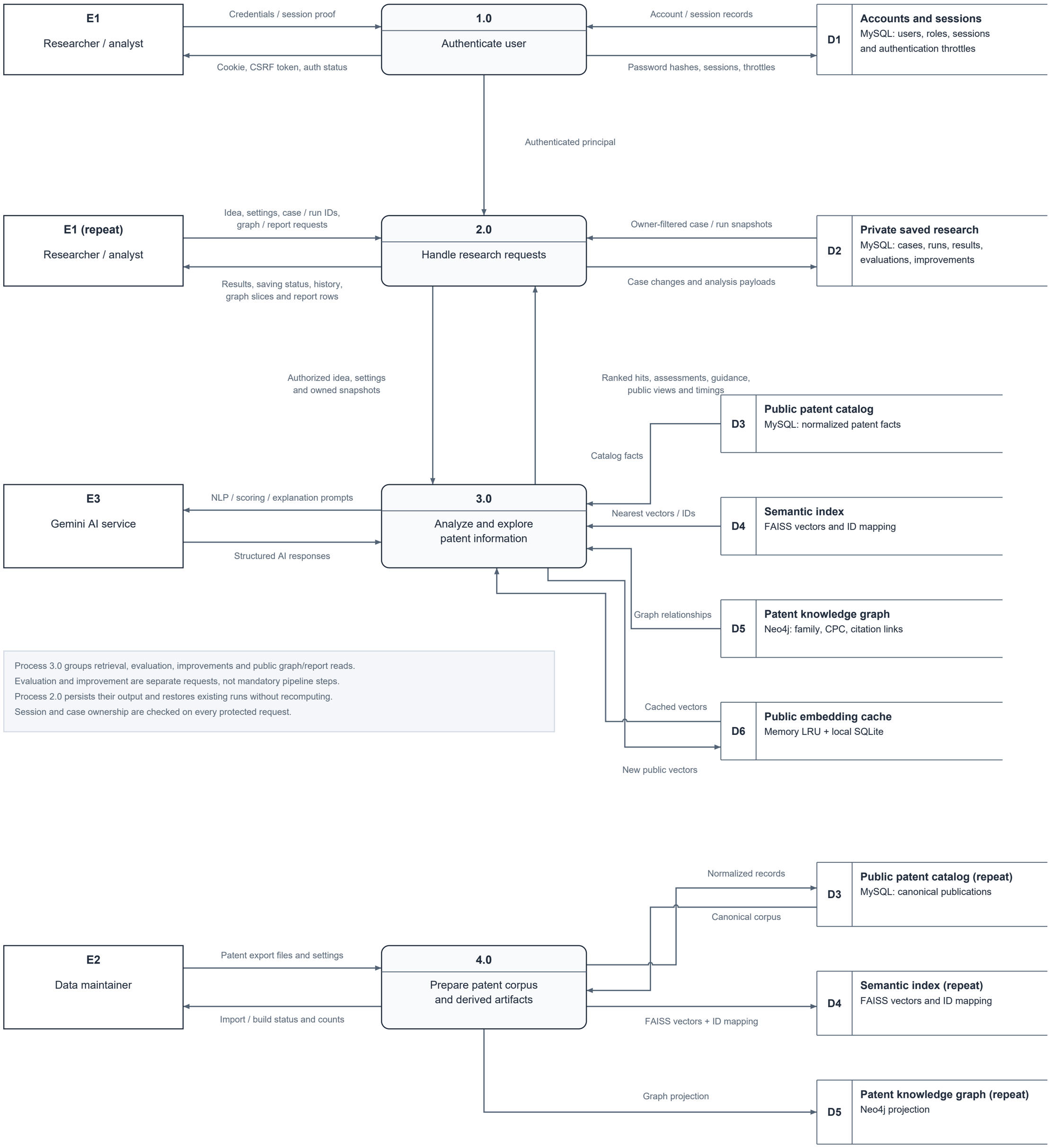
Rectangle: external entity | Numbered rounded box: process | Open-ended box: data store

Arrows name the data being transferred. Repeated IDs refer to the same entity/store. Line jumps are unconnected crossings.

PATENT INTELLIGENCE PLATFORM

Level 1 - Application processes and data stores

MySQL is authoritative. Neo4j and FAISS are derived projections; the embedding cache contains public patent vectors only.



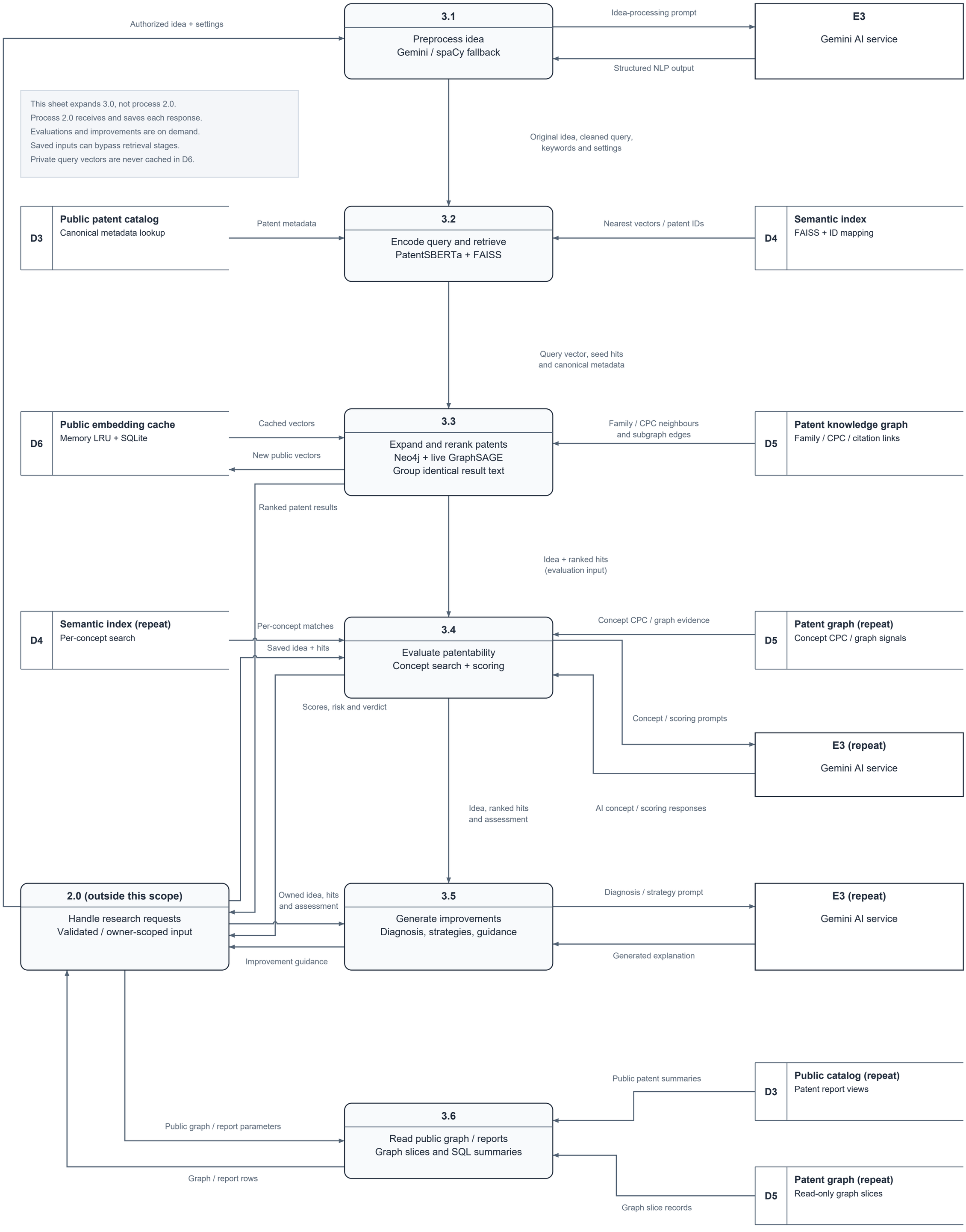
Rectangle: external entity | Numbered rounded box: process | Open-ended box: data store

Arrows name the data being transferred. Repeated IDs refer to the same entity/store. Line jumps are unconnected crossings.

PATENT INTELLIGENCE PLATFORM

Level 2 - Decomposition of process 3.0

Request-driven data transformations: retrieval, graph ranking, optional assessment/guidance, and read-only public exploration.



Rectangle: external entity | Numbered rounded box: process | Open-ended box: data store

Arrows name the data being transferred. Repeated IDs refer to the same entity/store. Line jumps are unconnected crossings.